Experiment No. 06

Aim: Verification of Kirchhoff's Currant law.

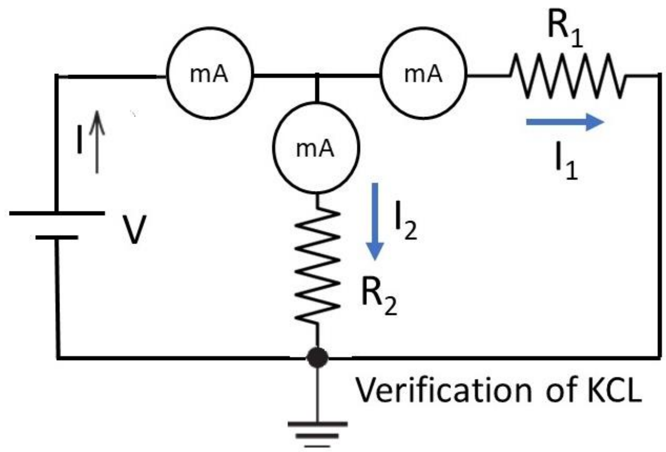
APPARATUS REQUIRED: 30V dc supply, Digital Multimeter as a voltmeter, Resistance Connecting Wires.

THEORY:

Node or junction in an electrical circuit is a point at which the number of branches meets. Kirchhoff's First law states that in a dc circuit the algebraic sum of all the currents at a junction is zero. The word algebraic indicates that the direction of currents is taken into account. The convention for the direction adopted for the current is that all currents flowing towards the junction are considered positive while the currents flowing away from the junction are taken as negative or vice-versa.

CIRCUIT DIAGRAM:

$$I = I_1 + I_2$$



PROCEDURE:

1. Make the connection according to the ckt diagram
2. Set the three rheostats to their max value.
3. Switch on the power supply
4. Change the setting of the rheostats to get different readings in all the ammeters.
5. Measure the current in the three ammeters
6. Check that at every time current in the main branch is equal to the sum of currentsin the two branches. repeat the setting of the rheostat
7. Switch off the power supply.

OBSERVATION TABLE:

KCL:

S.NO	Voltage (V)	Theoretical			Practical		
		I 1	I 2	I=I1+I2 (mA)	I1	I2	I=I1+I2 (mA)
1							
2							
3							

CALCULATION:

R total = R1||R2

R total = (R1*R2) / (R1+R2) = Ω

I=V/R = mA

I1 = I*(R2)/(R1+R2) = mA

I2=I*(R1)/(R1+R2) = mA

I = I1+I2 = mA

RESULT: The incoming current is found to be equal to the outgoing current

PRECAUTIONS:

1. Avoid loose connections.
2. Keep all the knobs in minimum position while switch on and off of the supply.
3. Input voltage increases minimum range.
4. All connections should be tight and correct.
5. Switch off the supply when not in use.
6. Reading should be taken carefully.

VIVA QUESTIONS:

1. What is another name for KCL?
2. Define network and circuit?
3. What is the property of inductor and capacitor?